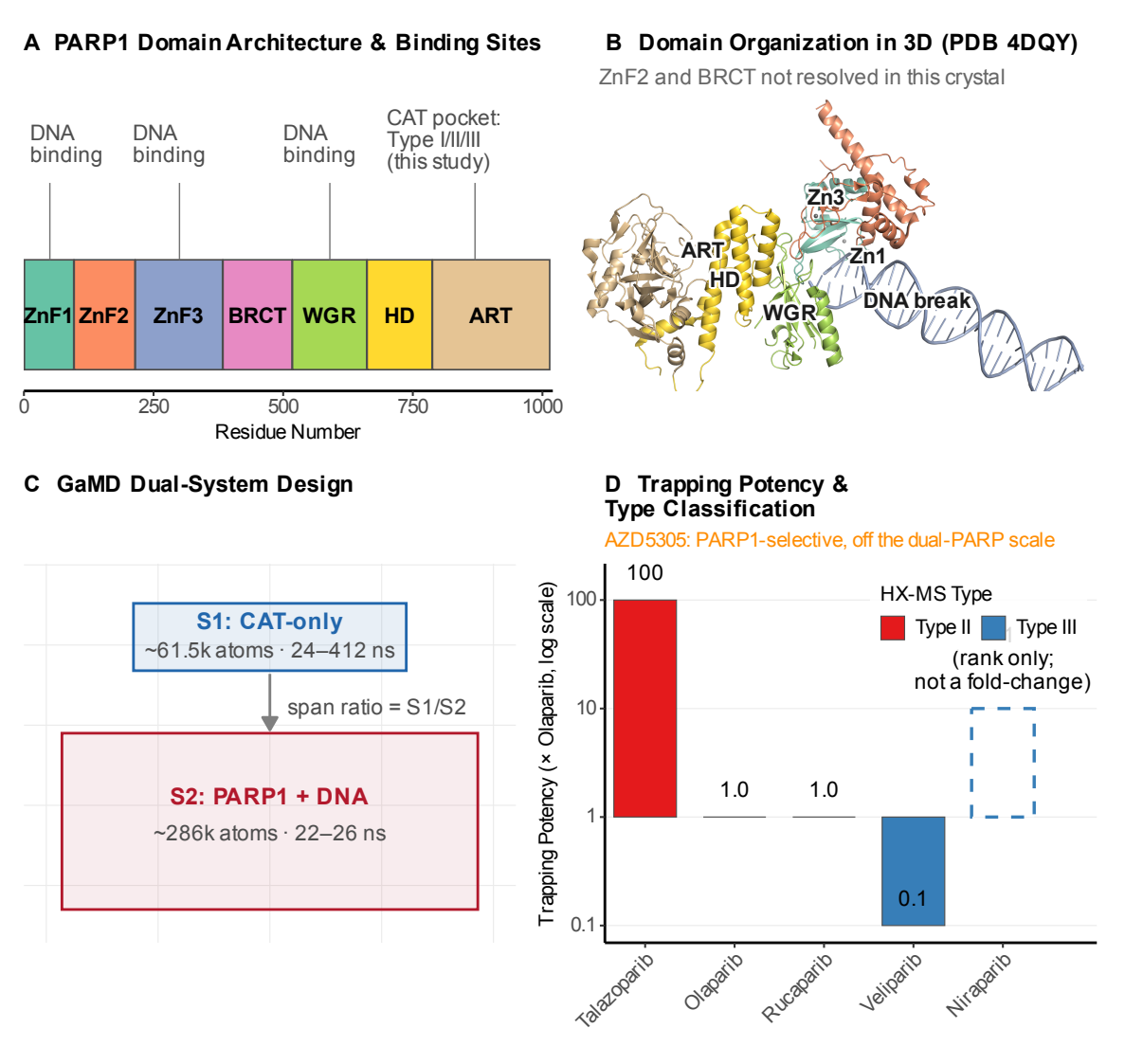

A PARP1 Domain Architecture & Binding Sites
DNA binding
DNA binding
DNA binding
CAT pocket: Type I/II/III (this study)
ZnF1
ZnF2
ZnF3
BRCT
WGR
HD
ART
0
250
500
750
1000
Residue Number
B Domain Organization in 3D (PDB 4DQY)
ZnF2 and BRCT not resolved in this crystal
Zn3
Zn1
ART
HD
WGR
DNA break
C GaMD Dual-System Design
S1: CAT-only
~61.5k atoms · 24–412 ns
span ratio = S1/S2
S2: PARP1 + DNA
~286k atoms · 22–26 ns
D Trapping Potency & Type Classification
AZD5305: PARP1-selective, off the dual-PARP scale
Trapping Potency (× Olaparib, log scale)
100
100
10
1
0.1
HX-MS Type
Type II
Type III
(rank only; not a fold-change)
1.0
1.0
0.1
Talazoparib
Olaparib
Rucaparib
Veliparib
Niraparib